THE DRAWN LENGTH OF THE CRITICAL BETA-SPLITTING TREE GROWS LINEARLY

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ABSTRACT. In the critical beta-splitting model of a random binary tree with n leaves, a clade of m leaves splits into clades of i and $m-i$ leaves with probability proportional to $1/(i(m-i))$. Aldous and Janson asked for a proof that the expected drawn length L_n of the cladogram of this tree grows roughly like $n\log^4 n$. We show that L_n/n converges to the finite constant $\lambda = \sum_{m\geq 2} a(m)c_m/m$, where $a(m) = 6h_{m-1}/(\pi^2(m-1))$ is the limiting probability that a uniformly chosen leaf lies in a clade of exactly m leaves and c_m is the expected contribution of the root of a tree with m leaves to the drawn length. Numerically $\lambda \approx 4.88$, and $4.80 < \lambda < 5.38$. The proof combines an exponential moment bound for the height with a bound, uniform in n, on the time the harmonic descent chain spends in a dyadic block. A deterministic comparison with the sum of the leaf depths gives the weaker bound $L_n = O(n\log^2 n)$ directly. The prediction $n\log^4 n$ comes from a heuristic in which the expected number of clades of size m is taken to be $na(n,m)$ instead of $(n/m)a(n,m)$.

1. INTRODUCTION

Let $h_k = \sum_{j=1}^{k} 1/j$ denote the harmonic numbers. The critical beta-splitting tree T_n, denoted DTCS(n) in [2], is the random rooted binary tree with n leaves constructed as follows. The root clade is the set of all n leaves. A clade of $m \geq 2$ leaves is split into a left clade of i leaves and a right clade of $m-i$ leaves with probability

$$q(m,i) = \frac{m}{2h_{m-1}\, i(m-i)}, \qquad 1 \leq i \leq m-1, \tag{1.1}$$

independently of all other splits, and a clade of one leaf is a leaf of the tree. This is the case $\beta = -1$ of the beta-splitting family of Aldous [1], and its structure has been studied in detail in recent years [4, 2, 3, 5, 6].

For a finite rooted binary tree T we write $h(T)$ for its height, the largest number of edges on a path from the root to a leaf, so that a single leaf has height 0, and T_v for the subtree consisting of a vertex v and its descendants. In the usual drawing of T as a cladogram, the leaves sit at level 0, an internal vertex v sits at level $h(T_v)$, and v is joined to each of its two children by a vertical line. The *drawn length* $\ell(T)$ is the total length of these vertical lines. If T_L and T_R are the subtrees of the two children of the root, then $h(T) = 1 + \max(h(T_L), h(T_R))$ and

$$\ell(T) = \ell(T_L) + \ell(T_R) + 2 + |h(T_L) - h(T_R)|, \tag{1.2}$$

with $\ell(T) = 0$ for a single leaf; this is the recursion (73) of [2]. Put

$$L_n = \mathbb{E}\,\ell(T_n), \qquad H_n = \mathbb{E}\,h(T_n).$$

In [2, Section 6.1.1] Aldous and Janson give a heuristic argument that L_n is of order $n\log^4 n$ and pose the following problem.

Open Problem 14 of [2]**.** Prove that L_n grows roughly like $n\log^4 n$.

Date: 12 October 2026.

2020 *Mathematics Subject Classification.* 60C05, 05C05, 60J10, 92B10.

Key words and phrases. Critical beta-splitting tree, cladogram, drawn length, harmonic descent chain, occupation measure.

The order is in fact n. For $m \geq 2$ let

$$c_m = \mathbb{E}\big[2 + |h(T_L) - h(T_R)|\big], \tag{1.3}$$

where T_L and T_R are the subtrees of the two children of the root of T_m, and let

$$a(m) = \frac{6h_{m-1}}{\pi^2(m-1)}, \qquad m \geq 2. \tag{1.4}$$

Theorem 1.1. *As $n \to \infty$,*

$$\frac{L_n}{n} \longrightarrow \lambda := \sum_{m=2}^{\infty} \frac{a(m)\, c_m}{m}.$$

The series converges, and $2 < \lambda < \infty$. Numerically $4.800 < \lambda < 5.372$, and $\lambda \approx 4.88$.

The numerical statements are explained in Section 5. A weaker upper bound, which already rules out the order $n \log^4 n$, follows from a deterministic comparison. Let $S(T)$ be the sum of the depths of the leaves of T (the Sackin index of T).

Proposition 1.2. *For every finite rooted binary tree T with n leaves, $2(n-1) \leq \ell(T) \leq S(T)$. Consequently*

$$2(n-1) \leq L_n \leq n\,\mathbb{E}D_n = \Big(\frac{3}{\pi^2} + o(1)\Big) n \log^2 n,$$

where D_n is the depth of a uniformly chosen leaf of T_n.

The first statement is part of Lemma 2.1 below, $\mathbb{E}S(T_n) = n\mathbb{E}D_n$ holds by averaging over the leaves, and $\mathbb{E}D_n \sim (3/\pi^2)\log^2 n$ is proved by Aldous and Pittel [4] (see also [2, Theorem 3]).

The heuristic of [2, (75)] starts from the expected number of clades of size m in T_n, which is $(n/m)a(n,m)$, where $a(n,m)$ is the probability that a uniformly chosen leaf lies in a clade of exactly m leaves; this identity is stated in [2, Section 4.8], and we prove it in Lemma 2.2. The first line of (75) uses $n\,a(n,m)$ instead. The two counts differ in total: $\sum_{m=2}^{n}(n/m)a(n,m) = n-1$ is the number of internal vertices of T_n, while $\sum_{m=2}^{n} n\,a(n,m) = n\mathbb{E}D_n$. With the factor $1/m$ in place, the same computation leads to the convergent series of Theorem 1.1.

The proof of Theorem 1.1 rests on two elementary estimates. Lemma 3.1 bounds $H_n \leq 4h_{n-1}\log n$ through an exponential moment of D_n, and Lemma 3.2 bounds the expected number of visits of the harmonic descent chain to a block $\{b+1,\dots,2b\}$ by $2h_{2b-1}$, uniformly in the starting point. Together with the convergence $a(n,m) \to a(m)$ for each fixed m, which is Theorem 18 of [2], they give the convergence of $L_n/n = \sum_m a(n,m)c_m/m$ by dominated convergence. The convergence of $\sum_m a(n,m)f(m)$ to $\sum_m a(m)f(m)$ for nonnegative f with $f(m) = O(m^{-\varepsilon})$ is a case of Ansatz 20 of [2] stated by Brandenberger, Chin and Mossel [5, Section 1.1], who also obtain rates for $a(n,m) \to a(m)$. Since $c_m/m = O(m^{-1/2})$ by Lemma 3.1, Theorem 1.1 also follows from their statement; the argument given here uses only the convergence $a(n,m) \to a(m)$. Ma [6] proved that $h(T_n)/\log^2 n$ converges almost surely and in L^p to an explicit constant $C_{\mathrm{ht}} \approx 0.976$, which answers Open Problem 4 of [2]; the cruder Lemma 3.1 is all that is needed here. Additive shape parameters with explicit tolls, such as the Sackin and Colless indices, are studied across the beta-splitting family in [7].

Theorem 1.1 is a statement about the mean. Whether $\ell(T_n)/n$ converges to λ in probability, and what the fluctuations are, we leave open, as well as Open Problem 15 of [2] on the profile of the drawn width.

Section 2 collects the identities, Section 3 proves the two estimates, Section 4 proves Theorem 1.1, and Section 5 describes the computations, including a Lean proof of Lemma 2.1.

2. Identities

For a finite rooted binary tree T let $n(T)$ be its number of leaves and $I(T)$ its set of internal vertices, so that $|I(T)| = n(T) - 1$.

Lemma 2.1. *Let T be a finite rooted binary tree with n leaves. Then*

(a) $\ell(T) = \sum_{v \in I(T)} h(T_v) + h(T)$;
(b) $S(T) = \sum_{v \in I(T)} n(T_v)$;
(c) $h(T) \le n - 1$;
(d) $2(n-1) \le \ell(T) \le S(T)$.

Proof. (a) The vertical line from an internal vertex v to a child u has length $h(T_v) - h(T_u)$. In the sum over all vertical lines, every internal vertex appears twice as a parent, every vertex other than the root appears once as a child, and leaves have height 0. Hence $\ell(T) = 2\sum_{v \in I(T)} h(T_v) - \sum_{v \in I(T) \setminus \{\text{root}\}} h(T_v)$, which is (a). (b) The depth of a leaf is the number of its internal ancestors, so each $v \in I(T)$ is counted once for each of the $n(T_v)$ leaves below it. (c) By induction, $h(T) = 1 + \max(h(T_L), h(T_R)) \le \max(n(T_L), n(T_R)) \le n - 1$. (d) By (1.2) every internal vertex contributes at least 2 to $\ell(T)$, which gives the lower bound. By (a), (c) applied to each T_v, and (b),

$$\ell(T) \le \sum_{v \in I(T)} \big(n(T_v) - 1\big) + (n-1) = S(T) - |I(T)| + (n-1) = S(T). \qquad \square$$

Fix n and let V be a leaf of T_n chosen uniformly at random. The sizes of the clades containing V, read from the root down, form a decreasing sequence $n = J_0 > J_1 > \cdots > J_{D_n} = 1$, where D_n is the depth of V. Given $J_t = k \ge 2$, the clade of size k splits as in (1.1) and V, being uniform in it, falls into a part of size j with probability j/k; since $q(k,j) = q(k,k-j)$,

$$\mathbb{P}(J_{t+1} = j \mid J_t = k) = 2q(k,j)\frac{j}{k} = \frac{1}{h_{k-1}(k-j)}, \qquad 1 \le j \le k-1. \tag{2.1}$$

So (J_t) is the harmonic descent chain of [3] started at n and stopped at 1 (see also [2, Section 3.1]). For $1 \le m \le n$ let $a(n,m)$ be the probability that this chain visits m, and put $a(n,m) = 0$ for $m > n$. Then $\mathbb{E}D_n = \sum_{m=2}^{n} a(n,m)$.

Lemma 2.2. *Let $N_n(m)$ be the number of internal vertices v of T_n with $n(T_v) = m$. Then $\mathbb{E}N_n(m) = (n/m)\, a(n,m)$ for $2 \le m \le n$.*

Proof. Two clades of the same size are either equal or disjoint, so given T_n the probability that V lies in a clade of size m is $mN_n(m)/n$. Taking expectations gives $a(n,m) = m\mathbb{E}N_n(m)/n$. $\square$

Lemma 2.3. *For $n \ge 2$,*

$$L_n = \sum_{m=2}^{n} \mathbb{E}N_n(m)\, c_m = n \sum_{m=2}^{n} \frac{a(n,m)\, c_m}{m}, \tag{2.2}$$

$$L_n = n \sum_{m=2}^{n} \frac{a(n,m)\, H_m}{m} + H_n. \tag{2.3}$$

Moreover $c_m = 2H_m - 2\sum_{i=1}^{m-1} q(m,i) H_i$ and $2 \le c_m \le 1 + H_m$ for $m \ge 2$.

Proof. Unrolling (1.2) gives $\ell(T) = \sum_{v \in I(T)} g(T_v)$ with $g(T') = 2 + |h(T'_L) - h(T'_R)|$. Conditionally on the splits outside it, the subtree below a clade of size m is distributed as T_m. Hence $\mathbb{E}\sum_{v \in I(T_n)} g(T_v) = \sum_m \mathbb{E}N_n(m)\, \mathbb{E}g(T_m) = \sum_m \mathbb{E}N_n(m) c_m$, and Lemma 2.2 gives (2.2). The same argument applied to Lemma 2.1(a) gives (2.3). For the formula for c_m use $|x-y| = 2\max(x,y) - x - y$, $\mathbb{E}\max(h(T_L), h(T_R)) = H_m - 1$, and $\mathbb{E}h(T_L) = \mathbb{E}h(T_R) = \sum_i q(m,i) H_i$, which holds because $q(m,i) = q(m,m-i)$. The lower bound $c_m \ge 2$ is clear, and $|x-y| \le \max(x,y)$ for $x, y \ge 0$ gives $c_m \le 2 + H_m - 1$. $\square$

Since $\sum_{m=2}^{n} \mathbb{E}N_n(m) = n-1$, Lemma 2.2 also gives

$$\sum_{m=2}^{n} \frac{a(n,m)}{m} = \frac{n-1}{n}. \tag{2.4}$$

3. Two estimates

Lemma 3.1. *For $n \geq 2$, $H_n \leq 4h_{n-1}\log n$. Consequently $c_m \leq 1 + 4h_{m-1}\log m$ for $m \geq 2$.*

Proof. Let $\theta = 1/(2h_{n-1})$. For $1 \leq k \leq n$ let $F(k) = \mathbb{E}_k e^{\theta D}$, where D is the number of steps the chain (2.1) started at k needs to reach 1. We show $F(k) \leq k$ by induction. Clearly $F(1) = 1$. Let $2 \leq k \leq n$ and assume $F(j) \leq j$ for $j < k$. Since

$$\sum_{j=1}^{k-1} \frac{j}{h_{k-1}(k-j)} = \sum_{r=1}^{k-1} \frac{k-r}{h_{k-1}\, r} = k - \frac{k-1}{h_{k-1}},$$

we get $F(k) \leq e^{\theta} \sum_{j<k} \mathbb{P}(J_1 = j \mid J_0 = k)\, j = e^{\theta} k(1-x_k)$ with $x_k = (k-1)/(kh_{k-1})$. Now $x_k \geq 1/(2h_{k-1}) \geq \theta$ and $x_k < 1$, so $e^{\theta} \leq 1/(1-\theta) \leq 1/(1-x_k)$ and $F(k) \leq k$.

Pointwise $e^{\theta h(T_n)}$ is at most the sum of $e^{\theta d(v)}$ over the leaves v, where $d(v)$ is the depth of v, and the expectation of this sum is $n\mathbb{E}e^{\theta D_n} = nF(n) \leq n^2$. By Jensen's inequality $e^{\theta H_n} \leq \mathbb{E}e^{\theta h(T_n)} \leq n^2$, that is, $H_n \leq 2\log n/\theta = 4h_{n-1}\log n$. The bound on c_m follows from Lemma 2.3. □

Lemma 3.2. *For all $n \geq 2$ and $b \geq 1$,*

$$\sum_{m=b+1}^{2b} a(n,m) \leq 2h_{2b-1}.$$

Proof. The sum is the expected number of visits of the chain (2.1) started at n to $B = \{b+1, \dots, 2b\}$. The chain is strictly decreasing, so its visits to B are consecutive. From any $k \in B$ it jumps to $\{1, \dots, b\}$ with probability

$$\sum_{j=1}^{b} \frac{1}{h_{k-1}(k-j)} \geq \frac{b}{h_{k-1}(k-1)} \geq \frac{b}{(2b-1)h_{2b-1}} \geq \frac{1}{2h_{2b-1}}.$$

Hence the number of visits to B is stochastically dominated by a geometric random variable with mean $2h_{2b-1}$. □

4. Proof of Theorem 1.1

Let $f(m) = c_m/m$. For $b \geq 1$ and $b < m \leq 2b$, Lemma 3.1 gives $f(m) \leq (1 + 4h_{2b-1}\log(2b))/b$, and with Lemma 3.2, for every n,

$$\sum_{m=b+1}^{2b} a(n,m) f(m) \leq \beta_b := \frac{2h_{2b-1}(1 + 4h_{2b-1}\log(2b))}{b}. \tag{4.1}$$

Since $h_{2^{k+1}-1} \leq 1 + (k+1)\log 2$, the numbers β_{2^k} are of order $k^3 2^{-k}$, and $\varepsilon_K := \sum_{k \geq K} \beta_{2^k}$ is finite and tends to 0 as $K \to \infty$. The blocks $\{2^k + 1, \dots, 2^{k+1}\}$, $k \geq 0$, partition $\{2, 3, \dots\}$, so (4.1) gives

$$\sum_{m > 2^K} a(n,m) f(m) \leq \varepsilon_K \qquad \text{for all } n \text{ and } K. \tag{4.2}$$

Each block sum in (4.1) has finitely many terms, so letting $n \to \infty$ and using $a(n,m) \to a(m)$ [2, Theorem 18] gives $\sum_{b<m\leq 2b} a(m)f(m) \leq \beta_b$, and hence $\sum_{m>2^K} a(m)f(m) \leq \varepsilon_K$. In particular $\lambda = \sum_m a(m)f(m) < \infty$. By (2.2) and (4.2),

$$\Big|\frac{L_n}{n} - \lambda\Big| \leq \sum_{m=2}^{2^K} |a(n,m) - a(m)|\, f(m) + 2\varepsilon_K.$$

The finite sum tends to 0 as $n \to \infty$, and then $\varepsilon_K \to 0$ as $K \to \infty$. Hence $L_n/n \to \lambda$.

For the lower bound, (2.2) and (2.4) give

$$\frac{L_n}{n} = \frac{2(n-1)}{n} + \sum_{m=2}^{n} \frac{a(n,m)(c_m - 2)}{m}.$$

Since $0 \le (c_m - 2)/m \le f(m)$, the argument above applies to the last sum, and therefore $\lambda = 2 + \sum_{m\ge 2} a(m)(c_m - 2)/m \ge 2 + a(4)(c_4 - 2)/4$. A tree with 4 leaves splits into parts of sizes 1 and 3 with probability 8/11, and then $|h(T_L) - h(T_R)| = 2$; otherwise both parts have height 1. So $c_4 = 2 + 16/11$ and $\lambda > 2$. The numerical bounds are proved in Section 5. □

Remark 4.1. By (2.3), Lemma 3.1, which gives $H_n/n \to 0$, and the same argument with H_m/m in place of $f(m)$, also

$$\lambda = \sum_{m=2}^{\infty} \frac{a(m)\, H_m}{m}.$$

The two series for λ have different terms; their sums agree because $c_m = 2H_m - 2\sum_i q(m,i)H_i$ and $\sum_{m>i} a(m)q(m,i)/m = a(i)/(2i)$, the limit form of the recursion $\mathbb{E}N_n(i) = \mathbf{1}\{i = n\} + 2\sum_{m>i} \mathbb{E}N_n(m)q(m,i)$.

5. Numerical values

The script `code/check.py` in the repository of the paper computes the distribution of $h(T_m)$ for $m \le 4000$ from the recursion

$$\mathbb{P}(h(T_m) \le x) = \sum_{i=1}^{m-1} q(m,i)\, \mathbb{P}(h(T_i) \le x-1)\, \mathbb{P}(h(T_{m-i}) \le x-1),$$

truncated at $x = 400$, beyond which the remaining probability for $m = 4000$ is below 10^{-12}. From it, it computes H_m, then c_m by Lemma 2.3, and L_m by the recursion $L_m = 2\sum_i q(m,i)L_i + c_m$, which follows from (1.2) and the branching property. For $n \in \{10, 100, 1000, 2000, 4000\}$ it also computes the expected clade counts from the recursion $\mathbb{E}N_n(m) = \mathbf{1}\{m = n\} + 2\sum_{k>m} \mathbb{E}N_n(k)q(k,m)$ and the probabilities $a(n,m)$ from the chain (2.1), and checks Lemma 2.2, which holds to a relative error below $3 \cdot 10^{-15}$, and (2.2): the two computations of L_n/n agree to within 10^{-9}. In exact rational arithmetic, the script computes the joint law of $(h(T_m), \ell(T_m))$ for $m \le 8$ by conditioning on the first split and checks Lemma 2.2 for $n \le 12$ and (2.2) for $n \le 8$; Lemma 2.1 is checked on all 6918 ordered binary trees with at most 10 leaves. Lemma 3.1 holds for $n \le 4000$ with largest ratio $H_n/(4h_{n-1}\log n)$ equal to 0.361, and Lemma 3.2 holds for $n = 4000$ and every $b \le 2000$ with largest ratio 0.407. Tables 1 and 2 show some of the values.

m	2	3	4	5	6	7
H_m	1	2	$\frac{30}{11}$	$\frac{189}{55}$	$\frac{30093}{7535}$	$\frac{239091}{52745}$
L_m	2	5	$\frac{90}{11}$	$\frac{646}{55}$	$\frac{2103}{137}$	$\frac{201977}{10549}$

Table 1. Exact mean height and mean drawn length of T_m.

The constant λ. All terms of the series for λ are positive, so the partial sum $\sum_{m=2}^{4000} a(m)c_m/m = 4.80059$ is a lower bound. For $m > 4000$ we use $c_m \le 1 + 4h_m \log m$ from Lemma 3.1; the resulting series is summed numerically up to $m = 10^8$ and bounded by an integral beyond, and its value is less than 0.5712. Hence $4.800 < \lambda < 5.372$. For $1000 \le m \le 4000$ the values c_m are fitted by $0.5756\log^2 m - 0.5604\log m + 4.7753$ with largest error $6 \cdot 10^{-4}$, and using this fit for $m > 4000$ gives $\lambda \approx 4.877$. The series of Remark 4.1, with H_m fitted in the same way, gives 4.877 as well,

n	H_n	L_n/n
10	5.8603	3.0988
100	18.5362	4.4286
1000	39.1932	4.7877
2000	47.1006	4.8239
4000	55.8147	4.8458

TABLE 2. Mean height and mean drawn length per leaf.

although its partial sum (4.7686) and its estimated remainder (0.1085) are different. The values of L_n/n in Table 2 increase slowly towards λ; by Lemma 3.1 the remainder $\sum_{m>n} a(m)c_m/m$ is at most of order $(\log n)^3/n$.

Lean. The file `lean/DrawnLength.lean` proves Lemma 2.1 for all finite rooted binary trees in Lean 4, using the core library only. The theorems depend on the axioms `propext` and `Quot.sound`. The probabilistic statements are not formalized.

ACKNOWLEDGEMENTS AND FUNDING

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [8] and working together inside the Brain framework [10], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [9], the Math Brain repository [8] and the Brain framework repository [10].

DECLARATION OF COMPETING INTEREST

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. Open Problem 14 of [2] was one of the open problems in probability on which the project's agents worked in September 2026. On 6 September 2026 an OpenAI Codex agent wrote the comparison of Proposition 1.2, observed that the heuristic (75) of [2] omits the factor $1/m$, and gave an argument for the linear limit in the form of Remark 4.1, based on a height bound through halving steps and the block estimate of Lemma 3.2. On 8 September 2026 a Claude agent (Anthropic) wrote a Lean proof of the inequality $\ell(T) \le S(T)$ and estimated L_n/n by simulation. On 10 October 2026 an OpenAI Codex agent checked the argument again. On 12 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, gave the exponential moment proof of Lemma 3.1, wrote the proofs in their present form, searched the literature, and wrote the computer checks, the Lean file of Section 5 and the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code and the Lean file are available at `https://github.com/yspennstate/beta-splitting-drawn-length`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [8].

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